

M SHARJEEL ASHIR

DATA CENTER ENGINEER

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Energy and Data Center Systems Engineer specializing in Model-Based Design, formal verification, and DevOps. Experienced in developing MPC-based control and optimization solutions for DC UPS, microgrids, and bidirectional EV charging. Designed advanced energy management and orchestration solutions for DC data centers, improving reliability and enabling faster deployment. Skilled at translating customer requirements into system architectures, technical specifications, and acceptance criteria while delivering robust control solutions and collaborating with cross-functional stakeholders.

Professional Experience

Data Center and Secure Power Business - Schneider Electric

Grenoble, France

Data Centers Control Systems Engineer

July 2025 - Present

- Architecting next-generation 800V DC microgrid and DC UPS solutions for future data centers, translating customer requirements into end-to-end system architectures and technical specifications, securing €50M in R&D funding.
- Designed and implemented CI/CD pipelines (GitHub Actions, Docker, RaaS) and a scalable verification and validation framework for 200K+ LOC projects, reducing PLC code generation and validation time by up to 80%.
- Led Hardware-in-the-Loop (HIL) validation framework development, deploying controller software on Speedgoat real-time hardware for timing analysis, fault injection, and requirements-based verification, while training cross-functional teams on HIL testing methodologies.
- Developed AI-driven load smoothing algorithms for DC UPS systems to predict load fluctuations and optimize battery dispatch, enabling ancillary services (FCR, FRR), improving grid flexibility, and reducing peak demand by 18%.

AI Hub / Energy Simulation Team - Schneider Electric

Grenoble, France

Energy Simulation and Management Engineer

Nov 2024 - July 2025

- Engineered Model Predictive Control (MPC) algorithms in MATLAB/Simulink for bidirectional EV charging in microgrids, demonstrating technical and economic feasibility while reducing simulation time by 85% and maintaining 99.8% accuracy for real-time applications.
- Designed battery degradation prediction models using weighted Ah-throughput in optimization tool, extending battery lifespan by 2+ years in Vehicle-to-Building scenarios through optimized charging strategies.
- Collaborated with the Sustainability Research Institute (SRI) on energy market modeling for mobile storage systems, quantifying 20–30% residential energy savings using prediction algorithms and translating findings into actionable insights for sales and marketing teams.

K State Energy Systems Center and Black & Veatch

Manhattan, United States

Junior Energy Engineer

July 2022 - Aug 2023

- Developed requalification tests for second life EV battery application in microgrids, receiving UL1974 certification and commissioning 3 pilot projects.
- Designed a two-stage optimization-based battery management system, experimentally proving a 20% improvement in energy efficiency.
- Developed and deployed ML-based Arc Fault Detection (AFD) model on STM32 embedded systems, achieving 98.5% classification accuracy with less than 2 ms inference latency for real-time protection systems.

Education

IMT Atlantique (École Des Mines)

Nantes, France

M.S – Energy Engineering

Sept 2023 - Sept 2025

- CGPA = 3.85/4.00 (France Excellence for Climate Change Scholarship)

National University of Sciences and Technology

Islamabad, Pakistan

Bachelors in Mechanical Engineering

Sept 2019 - June 2022

- CGPA = 3.94/4.00 (Double Gold Medal)

Kansas State University

Manhattan, United States

Control Systems - Global UGRAD

Aug 2022 - Aug 2023

- CGPA = 4.00/4.00 (Dean's List Recipient)

Core Skills

Certifications

Data Centers Specialization, GitHub Actions Specialization, Stanford Machine Learning Certification

Controls & Simulation

Electrical and Physical Modelling, Digital Twin, SCADA, Industrial networking, PLC Codgen, SIL/HIL/MIL testing

Energy Systems

Power Management, DC UPS, Cooling Systems, Battery systems (BESS), Microgrid Design and Optimization

Project Management

Agile methodologies, JAMA/JIRA Requirements tracking, Market Research, Scenario modelling

Programming & Tools

Python, MATLAB/Simulink, C++, GitHub Actions, Docker, CI/CD pipelines, Machine Learning (OpenCV, PyTorch)

Languages

English: C2 (Bilingual), French: B2 (Intermediate)

Projects

Optimization of placement and sizing of Batteries in power networks : Python and Pulp based approach

Developed linear programming model in Python/PuLP for optimal PV and battery sizing, raising renewable share to 90.86% and cutting costs by 56%.

Machine Learning-Based Fault Recognition for Industrial Compressors

Developed an LSTM model for compressor fault detection via acoustic signals, optimized for STM32F7xx microcontrollers, achieving 98.7% accuracy.